CHAPTER -1
SOUND WAVES

1. Which of the following statements are correct?

i) Sound is a transverse wave.
ii) Sound is a longitudinal wave.
iii) Sound is a longitudinal wave and light is a transverse wave.
iv) Sound is a transverse wave and light is a longitudinal wave.

a) i & ii b) i,ii ,iii & iv c) ii only d) both ii & iii (1)

2. Statement 1 : When an object vibrates freely, it vibrates in its innate frequency. This is the natural frequency of that object.

Statement 2 :Change in length of the object is the only factor, that will affect the natural frequency of an object.

a) Only statement 1 is correct
b) Only statement 2 is correct
c) Both statements are wrong
d) Both statements are correct (1)

3.The number of oscillations in one second is called frequency then, choose the correct SI Unit of frequency

(cycles per second, Hz, metre per second, metre) (1)

4.Among the following waves, choose the waves that do not require a medium for transmission

i) Radio waves ii) Seismic waves iii) Light waves iv) Sound waves

a) i & iii b) iv only c) iii only d) both ii & iii (1)

5. Statement : Echo is the sound heard after a while due to the reflection of the initial sound.
Reason : For the echo to be heard, the reflecting surface must be at least 17.5 m away

a) The statement and the reason are correct, the reason explains the statement.
b) The statement and the reason are correct, the reason does not explain the statement.
c) The statement is correct and the reason is wrong.
d) The statement is wrong and the reason is correct. (1)

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6. Explain the uses of ultrasonic waves in medical diagnosis and treatment. (2)

7. What measures can be taken to safeguard against tsunamis? (2)

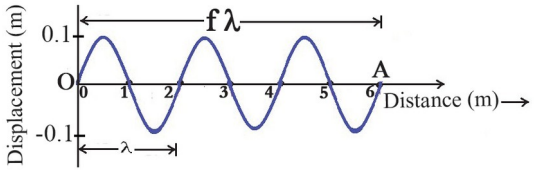
8.Can humans hear the sound of a Galton whistle? Justify your answer (2)

9. Explain the reason that the walls of large halls like cinema theatres made rough? (2)

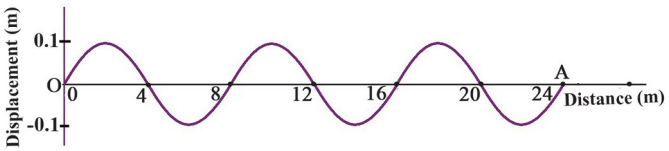
10. The echo of whistle sound is heard after 2 s by the person who whistled it. How far is the reflecting surface from the person hearing the echo? (speed of sound in air is 350 m/s). (2)

11. Name the devices in which multiple reflection of sound is utilized. (2)

12. Analyse the following two graphs



Graph 1



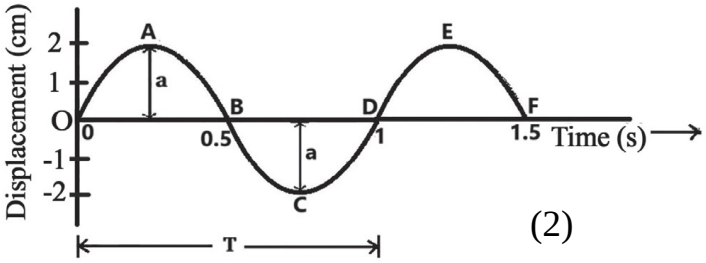
Graph 2

If the wave in the graph 1 takes 2s and graph 2 takes 8s to reach the point A, then compare the speed of both graphs and comment your statement. (2)

13. Observe the graph

a) Find out the time period, T of the wave

b) What is the amplitude of the wave



(2)

14.Some of the characteristics related to transverse waves and longitudinal waves are given below, Classify them

a) Particles in the medium vibrate perpendicular to the direction of propagation of the wave.

b) Compressions and rarefactions are formed.

c)Pressure variations occur in the medium.

d) Crests and troughs are formed.

e) Particles in the medium vibrate parallel to the direction of propagation of the wave.

f) No pressure variations occur in the medium. (3)

15. Depict the propagation of a transverse wave with a speed of 2 m/s, which travels to 10 m with a frequency of 0.5 Hz (3)

16. An electromagnetic wave with a speed of 3×10^8 m/s has a wave length of 750 nano metre (nm). Calculate its frequency. (3)

17.Explain the importance of Seismic Waves (3)

18.Two pith balls are suspended near the two prongs of a tuning fork fixed on a table so as to touch the prongs.

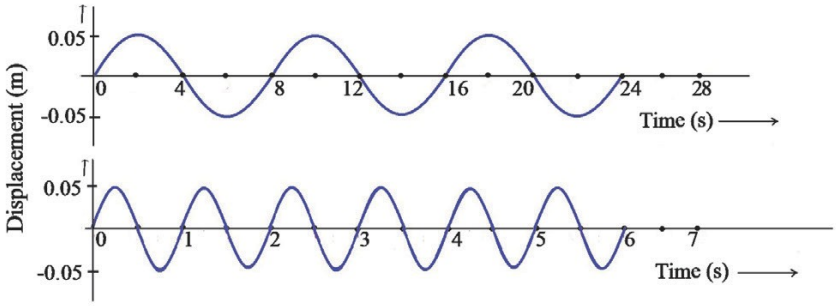
A person plays a piano sitting near this system.

a) In this case the pith balls move slightly. What may be the reason? (1)

b) While playing certain notes on the piano, the pith balls are thrown to a maximum distance. Which phenomenon is responsible for this? Explain (3)

19. A graphical illustration of two waves travelling at a speed of 3.2 m/s is given. Find out

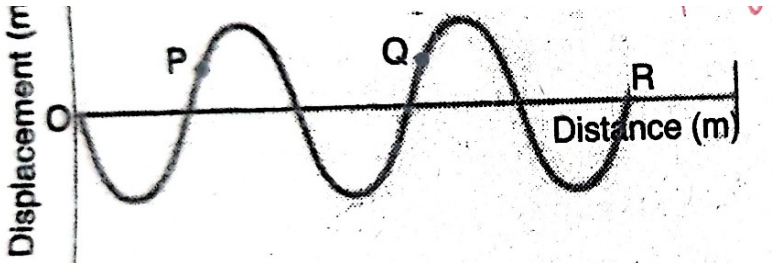
a) Frequency of the waves
b) Period of each wave
c)Wavelength of each wave.
d) Distance travelled by each wave



(4)

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20.Observe the figure



a) How many crests are there in the figure? How many troughs are there?

b) The wave takes 5 s to reach R from O. If so what is the frequency of this wave.

c) The point Q is at a distance of 2 m away form P. Calculate the speed of this wave.

d) Calculate the wavelength of this wave when the frequency is doubled without changing the speed. (4)

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ANSWER KEY

1.d
2.a
3.Hz
4.d
5.b

6.In the medical field, ultrasonic waves are used for diagnosis and treatment. To crush small stones in the kidneys and in physiotherapy also it used to take images of internal organs such as kidney, liver, gall bladder and uterus. Ultrasonic waves that travel through body tissues strike and reflect at areas of varying density in the tissues.

7.Early Warning Systems: Establishing and maintaining effective tsunami warning systems with sirens and communication channels.

Evacuation Plans: Developing and practising clear evacuation routes and procedures for coastal communities.

Coastal Defences: Constructing physical barriers like seawalls, breakwaters, and elevated land areas.

Education: Educating the public on the natural warning signs of a tsunami (e.g., sudden sea level drop) and how to respond.

8.No, The frequency of sound produced in a Galton whistle used for training dogs is about 30000 Hz. Humans cannot hear the sound of all frequencies. That is, there is a limit to the range of frequency of sounds that humans can hear. For a person with normal hearing, the lower limit of audible sound is about 20 Hz and the upper limit is about 20000 Hz (20 kHz).

9.The walls of large halls are made rough to reduce echoes and reverberation. Rough surfaces scatter sound waves in many different directions, rather than reflecting them uniformly like smooth surfaces. This scattering prevents the build-up of strong echoes and reduces the reverberation time, making the sound clearer and more intelligible for the audience.

10.Speed of sound = $\frac{\text{Total distance travelled}}{\text{Time}}$

$$v = \frac{2d}{t}$$

$$d = \frac{v \times t}{2}$$

$$= \frac{350 \times 2}{2}$$

$$= 350 \text{ m}$$

11. Soundboards, Curved ceilings in halls, Stethoscope etc

12. v = d/t , Graph 1- v = 6/2 = 3 m/s
Graph 2-v = 24/8 = 3 m/s
Both wave travel in same speed

13. a) Time period= 1 s

b) Amplitude = 2 cm

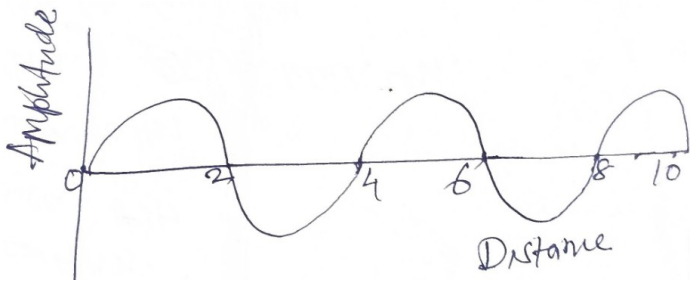
14. Transverse wave - a,d,f
Longitudinal wave - b,c,e

15. First find wavelength (λ) of wave

$$v = f \lambda$$

$$\lambda = v / f$$

$$\lambda = 2 / 0.5 = 20/5 = 4 \text{ m}$$



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16. The wavelength (λ) is given as 750 nanometres (nm).

$\lambda = 750nm = 750 \times 10^{-9} m$

The frequency (f) can be calculated using the wave speed equation, $c = f\lambda$, where c is the speed of light.

$$F = c / \lambda = \frac{3 \times 10^{8} m/S}{750 \times 10^{-9} m}$$

$$f = 4 \times 10^{14} \text{ Hz}$$

17.Seismic waves, generated by earthquakes or explosions, are vital for imaging Earth's deep interior, enabling scientists to identify layers like the crust, mantle, and core. By analysing wave velocity and refraction, they map structural composition, aid in locating natural resources, and enable, earthquake and tsunami risk assessment, saving lives.

18. a) Due to Forced vibration

b) Resonance, If the natural frequency of the forcing object and that of the forced object are equal, the objects are said to be in resonance. The objects undergoing resonance will vibrate with maximum amplitude.

19. a) $f_1 = 3/24 = 1/8 = 0.125$ Hz, $f_2 = 6/6 = 1$ Hz

b) $T_1 = 24/3 = 8$ s $T_2 = 6/6 = 1$ s

c) $\lambda_1 = v_1 / f_1 = 3.2 / 0.125 = 25.6$ m

$\lambda_2 = v_2 / f_2 = 3.2 / 1 = 3.2$ m

d) distance $d_1 = v_1 \times t_1 = 3.2 \times 24 = 76.8$ m

$d_2 = v_2 \times t_2 = 3.2 \times 6 = 19.2$ m

20. Crests – 2, Troughs - 3

b) $f = 2.5 / 5 = 1/2 = 0.5$ Hz

c) $\lambda = 2$ m , $v = f \times \lambda$ $v = 0.5 \times 2 = 1$ m/s

d) wavelength of this wave when the frequency is doubled without changing the speed.

$\lambda = v / f$, $\lambda = 1 / 1 = 1$ m

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